Trigonometry

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Right triangle trigonometric ratios**

In a right triangle, angle θ has an opposite side of 8 and hypotenuse 17. Find $\sin\theta$, $\cos\theta$, and $\tan\theta$.

The adjacent side: $\sqrt{17^2 - 8^2} = \sqrt{289 - 64} = \sqrt{225} = 15$. $\sin\theta = \frac{8}{17}$, $\cos\theta = \frac{15}{17}$, $\tan\theta = \frac{8}{15}$.

2 **[Foundational]** • **Target: Right triangle trigonometric ratios**

A ladder 15 feet long leans against a wall, making a 70° angle with the ground. Find the height reached on the wall, to the nearest tenth.

Height $= 15\sin(70^\circ) \approx 15(0.9397) \approx 14.1$ feet.

3 **[Foundational]** • **Target: The unit circle and radian measure**

Convert 150° to radians, in terms of π.

$150^\circ \times \frac{\pi}{180^\circ} = \frac{5\pi}{6}$.

4 **[Foundational]** • **Target: The unit circle and radian measure**

Find the exact value of $\sin\left(\frac{\pi}{6}\right)$ and $\cos\left(\frac{\pi}{4}\right)$, using the unit circle.

$\sin\left(\frac{\pi}{6}\right) = \frac{1}{2}$. $\cos\left(\frac{\pi}{4}\right) = \frac{\sqrt{2}}{2}$.

5 **[Foundational]** • **Target: Graphs of sine and cosine functions**

State the amplitude and period of $y = 3\sin(2x)$.

Amplitude: $|3| = 3$. Period: $\frac{2\pi}{2} = \pi$.

6 **[Foundational]** • **Target: Graphs of sine and cosine functions**

State the amplitude, period, and vertical shift of $y = 4\cos(x) + 2$.

Amplitude: 4. Period: 2π (unchanged, since there is no coefficient on x inside the cosine). Vertical shift: up 2 units.

7 **[Foundational]** • **Target: Law of Sines**

In triangle ABC, $A = 40^\circ$, $B = 65^\circ$, and $a = 10$. Find side b, to the nearest tenth, using the Law of Sines.

Law of Sines: $\frac{a}{\sin A} = \frac{b}{\sin B}$. $\frac{10}{\sin 40^\circ} = \frac{b}{\sin 65^\circ}$. $b = \frac{10\sin 65^\circ}{\sin 40^\circ} \approx \frac{10(0.9063)}{0.6428} \approx 14.1$.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Law of Cosines**

In triangle ABC, $a = 7$, $b = 9$, and $C = 60^\circ$. Find side c, using the Law of Cosines.

$c^2 = a^2 + b^2 - 2ab\cos C = 49 + 81 - 2(7)(9)\cos 60^\circ = 130 - 126(0.5) = 130 - 63 = 67$.
$c = \sqrt{67} \approx 8.2$.

9 **[Intermediate]** • **Target: Applications of trigonometry to real-world problems**

This question has two parts. A surveyor stands 200 feet from the base of a tower and measures the angle of elevation to the top as 38°. (a) Find the height of the tower, to the nearest foot. (b) If the surveyor moves 50 feet closer to the tower, find the new angle of elevation, to the nearest degree, using the height found in part (a).

(a) Height $= 200\tan(38^\circ) \approx 200(0.7813) \approx 156$ feet. (b) New distance: $200 - 50 = 150$ feet. New angle: $\theta = \arctan\left(\frac{156}{150}\right) = \arctan(1.04) \approx 46^\circ$.

10 **[Intermediate]** • **Target: Reciprocal trigonometric functions**

If $\sin\theta = \frac{5}{13}$, find $\csc\theta$.

$\csc\theta = \frac{1}{\sin\theta} = \frac{13}{5}$.

11 **[Intermediate]** • **Target: Reciprocal trigonometric functions**

If $\cos\theta = \frac{3}{5}$, find $\sec\theta$ and $\tan\theta$ (assuming θ is in the first quadrant).

$\sec\theta = \frac{1}{\cos\theta} = \frac{5}{3}$. Since $\sin^2\theta = 1 - \frac{9}{25} = \frac{16}{25}$, $\sin\theta = \frac{4}{5}$ (positive in the first quadrant).
$\tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{4/5}{3/5} = \frac{4}{3}$.

12 **[Intermediate]** • **Target: Angle of elevation and depression**

From the top of a 100-foot cliff, the angle of depression to a boat is 25°. Find the horizontal distance from the base of the cliff to the boat, to the nearest foot.

The angle of depression from the top equals the angle of elevation from the boat. Using the tangent ratio: $\tan(25^\circ) = \frac{100}{d}$, so $d = \frac{100}{\tan(25^\circ)} \approx \frac{100}{0.4663} \approx 214$ feet.

13 **[Intermediate]** • **Target: A multi-part triangle problem combining Law of Sines and Law of Cosines**

This question has two parts. Triangle ABC has $a = 8$, $b = 11$, and $C = 50^\circ$. (a) Use the Law of Cosines to find side c, to the nearest tenth. (b) Use the Law of Sines to find angle A, to the nearest degree.

(a) $c^2 = 8^2 + 11^2 - 2(8)(11)\cos 50^\circ = 64 + 121 - 176(0.6428) \approx 185 - 113.1 \approx 71.9$.
$c \approx \sqrt{71.9} \approx 8.5$. (b) $\frac{a}{\sin A} = \frac{c}{\sin C}$: $\frac{8}{\sin A} = \frac{8.5}{\sin 50^\circ}$. $\sin A = \frac{8\sin 50^\circ}{8.5} \approx \frac{8(0.766)}{8.5} \approx 0.721$, so $A \approx 46^\circ$.

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14 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the exact value of $\sin\left(\frac{\pi}{6}\right)$ and $\cos\left(\frac{\pi}{4}\right)$, using the unit circle.

$\sin\left(\frac{\pi}{6}\right) = \frac{1}{2}$. $\cos\left(\frac{\pi}{4}\right) = \frac{\sqrt{2}}{2}$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 7.01 Statistics & Probability

If $\sin\theta = \frac{5}{13}$, find $\csc\theta$.

$\csc\theta = \frac{1}{\sin\theta} = \frac{13}{5}$.

16 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 8.01 Matrices, Vectors & Complex Numbers

If $\cos\theta = \frac{3}{5}$, find $\sec\theta$ and $\tan\theta$ (assuming θ is in the first quadrant).

$\sec\theta = \frac{1}{\cos\theta} = \frac{5}{3}$. Since $\sin^2\theta = 1 - \frac{9}{25} = \frac{16}{25}$, $\sin\theta = \frac{4}{5}$ (positive in the first quadrant).
$\tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{4/5}{3/5} = \frac{4}{3}$.

17 [Challenging] • Target: The unit circle and radian measure • Cross-Lesson: 1.01 Numbers, Percents & Ratios

Convert $\frac{3\pi}{4}$ radians to degrees.

$\frac{3\pi}{4} \times \frac{180^\circ}{\pi} = 135^\circ$.

18 [Challenging] • Target: Basic trigonometric identities • Cross-Lesson: 2.01 Elementary & Intermediate Algebra

Use the Pythagorean identity to find $\cos\theta$, given that $\sin\theta = \frac{3}{5}$ and θ is in the first quadrant.

$\sin^2\theta + \cos^2\theta = 1$, so $\cos^2\theta = 1 - \frac{9}{25} = \frac{16}{25}$. Since θ is in the first quadrant, $\cos\theta$ is positive: $\cos\theta = \frac{4}{5}$.

19 [Challenging] • Target: Basic trigonometric identities • Cross-Lesson: 3.01 Functions & Graphs

Simplify $\frac{\sin\theta}{\cos\theta} \cdot \cos\theta$.

$\frac{\sin\theta}{\cos\theta} \cdot \cos\theta = \sin\theta$ (the $\cos\theta$ terms cancel).

20 [Challenging] • Target: Trigonometric equations • Cross-Lesson: 4.01 Coordinate Geometry

Solve $2\sin\theta = 1$ for $0^\circ \le \theta < 360^\circ$.

$\sin\theta = \frac{1}{2}$. Solutions in $[0^\circ, 360^\circ)$: $\theta = 30^\circ$ or $\theta = 150^\circ$ (both quadrants where sine is positive).

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Right triangle trigonometric ratios

In a right triangle, the angle θ has an opposite side of length 20, adjacent side 21, and hypotenuse 29. Find $\tan\theta$.

a. $\frac{29}{21}$

b. $\frac{20}{21}$ ✓

c. $\frac{29}{20}$

d. $\frac{21}{20}$

$\tan\theta = \textit{opposite}/\textit{adjacent} = 20/21 = \frac{20}{21}$.

22 [Foundational] • Target: The unit circle and radian measure

In a right triangle, the angle θ has an opposite side of length 9, adjacent side 12, and hypotenuse 15. Find $\sin\theta$.

a. $\frac{4}{3}$

b. $\frac{3}{5}$ ✓

c. $\frac{5}{4}$

d. $\frac{5}{3}$

$\sin\theta = \textit{opposite}/\textit{hypotenuse} = 9/15 = \frac{3}{5}$.

23 [Foundational] • Target: Basic trigonometric identities

In a right triangle, the angle θ has an opposite side of length 20, adjacent side 21, and hypotenuse 29. Find $\cos\theta$.

a. $\frac{29}{20}$

b. $\frac{21}{20}$

c. $\frac{21}{29}$ ✓

d. $\frac{29}{21}$

$\cos\theta = \textit{adjacent}/\textit{hypotenuse} = 21/29 = \frac{21}{29}$.

24 [Foundational] • Target: Graphs of sine and cosine functions

In a right triangle, the angle θ has an opposite side of length 9, adjacent side 12, and hypotenuse 15. Find $\cos\theta$.

a. $\frac{5}{4}$

b. $\frac{5}{3}$

c. $\frac{4}{5}$ ✓

d. $\frac{4}{3}$

$\cos\theta = \textit{adjacent}/\textit{hypotenuse} = 12/15 = \frac{4}{5}$.

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25 [Foundational] • Target: Law of Sines

In a right triangle, the angle θ has an opposite side of length 12, adjacent side 16, and hypotenuse 20. Find $\tan\theta$.

a. $\frac{4}{3}$

b. $\frac{3}{4}$ ✓

c. $\frac{5}{3}$

d. $\frac{5}{4}$

$\tan\theta = \textit{opposite}/\textit{adjacent} = 12/16 = \frac{3}{4}$.

26 [Foundational] • Target: Law of Cosines

In a right triangle, the angle θ has an opposite side of length 9, adjacent side 12, and hypotenuse 15. Find $\tan\theta$.

a. $\frac{5}{4}$

b. $\frac{5}{3}$

c. $\frac{3}{4}$ ✓

d. $\frac{4}{3}$

$\tan\theta = \textit{opposite}/\textit{adjacent} = 9/12 = \frac{3}{4}$.

27 [Foundational] • Target: Trigonometric equations

In a right triangle, the angle θ has an opposite side of length 12, adjacent side 16, and hypotenuse 20. Find $\cos\theta$.

a. $\frac{5}{4}$

b. $\frac{5}{3}$

c. $\frac{4}{5}$ ✓

d. $\frac{4}{3}$

$\cos\theta = \textit{adjacent}/\textit{hypotenuse} = 16/20 = \frac{4}{5}$.

28 [Foundational] • Target: Applications of trigonometry to real-world problems

In a right triangle, the angle θ has an opposite side of length 7, adjacent side 24, and hypotenuse 25. Find $\sin\theta$.

a. $\frac{7}{25}$ ✓

b. $\frac{25}{7}$

c. $\frac{24}{7}$

d. $\frac{25}{24}$

$\sin\theta = \textit{opposite}/\textit{hypotenuse} = 7/25 = \frac{7}{25}$.

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Reciprocal trigonometric functions**

Convert 170° to radians, in terms of π.

a. $\frac{18}{17}\pi$

b. $\frac{17}{18}\pi$ ✓

c. $\frac{10}{9}\pi$

d. $\frac{17}{9}\pi$

$170^\circ \times \frac{\pi}{180^\circ} = \frac{17}{18}\pi$.

30 **[Intermediate]** • **Target: Angle of elevation and depression**

Convert 350° to radians, in terms of π.

a. $\frac{35}{18}\pi$ ✓

b. $\frac{18}{35}\pi$

c. $\frac{19}{9}\pi$

d. $\frac{35}{9}\pi$

$350^\circ \times \frac{\pi}{180^\circ} = \frac{35}{18}\pi$.

31 **[Intermediate]** • **Target: Solving trigonometric equations over a restricted domain**

Convert 60° to radians, in terms of π.

a. $\frac{2}{3}\pi$

b. $\frac{1}{3}\pi$ ✓

c. $\frac{1}{2}\pi$

d. 3π

$60^\circ \times \frac{\pi}{180^\circ} = \frac{1}{3}\pi$.

32 **[Intermediate]** • **Target: A multi-part triangle problem combining Law of Sines and Law of Cosines**

Convert 210° to radians, in terms of π.

a. $\frac{4}{3}\pi$

b. $\frac{6}{7}\pi$

c. $\frac{7}{3}\pi$

d. $\frac{7}{6}\pi$ ✓

$210^\circ \times \frac{\pi}{180^\circ} = \frac{7}{6}\pi$.

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33 [Intermediate] • Target: Cofunction and negative angle identities

Convert 45° to radians, in terms of π.

a. $\frac{5}{12}\pi$

b. $\frac{1}{2}\pi$

c. 4π

d. $\frac{1}{4}\pi$ ✓

$45^\circ \times \frac{\pi}{180^\circ} = \frac{1}{4}\pi$.

34 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Convert 90° to radians, in terms of π.

a. π

b. $\frac{1}{2}\pi$ ✓

c. $\frac{2}{3}\pi$

d. 2π

$90^\circ \times \frac{\pi}{180^\circ} = \frac{1}{2}\pi$.

35 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Convert 15° to radians, in terms of π.

a. 12π

b. $\frac{1}{12}\pi$ ✓

c. $\frac{1}{4}\pi$

d. $\frac{1}{6}\pi$

$15^\circ \times \frac{\pi}{180^\circ} = \frac{1}{12}\pi$.

36 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Convert 160° to radians, in terms of π.

a. $\frac{9}{8}\pi$

b. $\frac{16}{9}\pi$

c. $\frac{19}{18}\pi$

d. $\frac{8}{9}\pi$ ✓

$160^\circ \times \frac{\pi}{180^\circ} = \frac{8}{9}\pi$.

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In triangle ABC, $A = 32^\circ$, $B = 57^\circ$, and $a = 9$. Find side b, to the nearest tenth, using the Law of Sines.

a. 14.2 ✓

b. 12.2

c. 16.2

d. 18.5

Law of Sines: $\frac{a}{\sin A} = \frac{b}{\sin B}$. $\frac{9}{\sin 32^\circ} = \frac{b}{\sin 57^\circ}$. $b = \frac{9\sin 57^\circ}{\sin 32^\circ} \approx \frac{9(0.8387)}{0.5299} \approx 14.2$.

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Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)** • **Cross-Lesson: 2.01 Elementary & Intermediate Algebra**

In triangle ABC, $A = 48^\circ$, $B = 34^\circ$, and $a = 5$. Find side b, to the nearest tenth, using the Law of Sines.

a. 3.8 ✓

b. 1.8

c. 5.8

d. 4.9

Law of Sines: $\frac{a}{\sin A} = \frac{b}{\sin B}$. $\frac{5}{\sin 48^\circ} = \frac{b}{\sin 34^\circ}$. $b = \frac{5\sin 34^\circ}{\sin 48^\circ} \approx \frac{5(0.5592)}{0.7431} \approx 3.8$.

39 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)** • **Cross-Lesson: 3.01 Functions & Graphs**

In triangle ABC, $A = 34^\circ$, $B = 36^\circ$, and $a = 17$. Find side b, to the nearest tenth, using the Law of Sines.

a. 17.9 ✓

b. 23.2

c. 15.9

d. 19.9

Law of Sines: $\frac{a}{\sin A} = \frac{b}{\sin B}$. $\frac{17}{\sin 34^\circ} = \frac{b}{\sin 36^\circ}$. $b = \frac{17\sin 36^\circ}{\sin 34^\circ} \approx \frac{17(0.5878)}{0.5592} \approx 17.9$.

40 **[Challenging]** • **Target: Right triangle trigonometric ratios** • **Cross-Lesson: 4.01 Coordinate Geometry**

In triangle ABC, $a = 10$, $b = 12$, and $C = 105^\circ$. Find side c, to the nearest tenth, using the Law of Cosines.

a. 17.5 ✓

b. 21.0

c. 16.0

d. 19.0

$c^2 = a^2 + b^2 - 2ab\cos C = 100 + 144 - 2(10)(12)\cos 105^\circ = 244 - 240(-0.2588) \approx 306.1$.
$c \approx \sqrt{306.1} \approx 17.5$.

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41 [Challenging] • Target: The unit circle and radian measure • Cross-Lesson: 5.01 Plane Geometry

In triangle ABC, $a = 6$, $b = 12$, and $C = 30^\circ$. Find side c, to the nearest tenth, using the Law of Cosines.

a. 5.9

b. 7.4 ✓

c. 7.9

d. 8.9

$c^2 = a^2 + b^2 - 2ab\cos C = 36 + 144 - 2(6)(12)\cos 30^\circ = 180 - 144(0.8660) \approx 55.3$.
$c \approx \sqrt{55.3} \approx 7.4$.

42 [Challenging] • Target: Basic trigonometric identities • Cross-Lesson: 7.01 Statistics & Probability

Solve $2\cos\theta - 1 = 0$ for $0^\circ \le \theta < 360^\circ$.

a. $270^\circ, 300^\circ$

b. $30^\circ, 210^\circ$

c. $45^\circ, 315^\circ$

d. $60^\circ, 300^\circ$ ✓

$\cos\theta = \frac{1}{2}$. Solutions in $[0^\circ, 360^\circ)$: $\theta = 60^\circ$ or $\theta = 300^\circ$.

43 [Challenging] • Target: Graphs of sine and cosine functions • Cross-Lesson: 8.01 Matrices, Vectors & Complex Numbers

Solve $\tan\theta + 1 = 0$ for $0^\circ \le \theta < 360^\circ$.

a. $135^\circ, 315^\circ$ ✓

b. $135^\circ, 240^\circ$

c. $60^\circ, 240^\circ$

d. $90^\circ, 300^\circ$

$\tan\theta = -1$. Solutions in $[0^\circ, 360^\circ)$: $\theta = 135^\circ$ or $\theta = 315^\circ$.

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.01 Numbers, Percents & Ratios

What is the exact value of $\cos(\frac{5\pi}{6})$?

a. $\frac{\sqrt{3}}{2}$

b. $-\frac{\sqrt{3}}{2}$ ✓

c. $\frac{1}{2}$

d. $-\frac{1}{2}$

$\frac{5\pi}{6}$ is in Quadrant II where cosine is negative. Reference angle is $\pi - \frac{5\pi}{6} = \frac{\pi}{6}$. Therefore, $\cos(\frac{5\pi}{6}) = -\cos(\frac{\pi}{6}) = -\frac{\sqrt{3}}{2}$.

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45 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.01 Elementary & Intermediate Algebra

In triangle PQR, sides $p = 5$, $q = 8$, and the included angle $R = 60^\circ$. What is the length of side r?

a. 6

b. 7 ✓

c. 8

d. $\sqrt{49}$

By Law of Cosines: $r^2 = 5^2 + 8^2 - 2(5)(8)\cos 60^\circ = 25 + 64 - 80(\frac{1}{2}) = 89 - 40 = 49$. Thus $r = 7$.

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